ELECTRON SPIN RESONANCE (ESR)

IBRAHIM H.I. ABUSHAWISH
Istanbul University, Department of Physics
Instructor:
Experiment Date: October 20, 2026 , Report Submission Date:
Course & Section Number: PHYS4704

***Abstract*—In this experiment, Electron Spin Resonance (ESR) was observed in a polycrystalline DPPH (1,1-diphenyl-2-picrylhydrazyl) sample using an RF frequency of $f = 146.0\,\mathrm{MHz}$ inside a pair of Helmholtz coils ($N = 241$, $R = 4.8\,\mathrm{cm}$). The resonance was observed on an oscilloscope in X-Y mode by applying a $50\,\mathrm{Hz}$ AC sweep modulation to the static magnetic field. Across three trials, the mean resonant current was measured as $\bar{I}_r = 1.250(6)\,\mathrm{A}$ ($s = 0.010\,\mathrm{A}$), corresponding to a resonant magnetic field of $\bar{B}_r = 5.088(23)\,\mathrm{mT}$ using the calibrated coil constant $k_B = 4.07 \times 10^{-3}\,\mathrm{T\,A^{-1}}$. This yields an experimental Landé factor of $g = 2.0521 \pm 0.0095$ ($s = 0.0164$), which has a relative error of $2.41\,\%$ compared to the literature value ($g_{\text{lit}} = 2.0037$) and $2.49\,\%$ compared to the free-electron value ($g_e = 2.0023$). The absorption linewidth was measured by finding the currents at the half-maximum points ($I_1 = 0.15\,\mathrm{A}, I_2 = 0.09\,\mathrm{A}$), giving $\Delta B = 0.2442\,\mathrm{mT}$ with a relative error of $12.79\,\%$ compared to $\Delta B_{\text{lit}} = 0.2800\,\mathrm{mT}$. The results confirm electron spin quantization and the magnetic dipole transition condition.**

I. Introduction

Electron spin was first proposed by Uhlenbeck and Goudsmit in 1925 to explain atomic fine structure and the anomalous Zeeman effect. When placed in an external magnetic field, the electron's magnetic moment aligns either parallel or antiparallel to the field, splitting the energy level into two discrete Zeeman sublevels. Electron Spin Resonance (ESR), discovered by Zavoisky in 1944, allows direct observation of transitions between these sublevels by applying a perpendicular radiofrequency (RF) magnetic field.

In this experiment, ESR is investigated using DPPH (1,1-diphenyl-2-picrylhydrazyl). In DPPH, the unpaired electron has its orbital angular momentum quenched by molecular bonding ($L \approx 0$). Its magnetic dipole moment arises purely from electron spin ($S = 1/2$), making DPPH an ideal standard sample for ESR calibration. The objectives of this experiment are: (1) to derive the resonance condition and the Helmholtz coil field equation, (2) to determine the Landé g-factor from repeated measurements, and (3) to measure the absorption linewidth (FWHM) of the resonance signal.

II. Theory

A. Magnetic Moments and Spin

An electron with spin angular momentum $\vec{S}$ has an intrinsic magnetic dipole moment:

$$\vec{\mu}_s = -g\frac{e}{2m_e}\vec{S} \tag{1}$$

Along the quantization axis (z), the spin projection is $S_z = m_s\hbar$ with $m_s = \pm 1/2$. In terms of the Bohr magneton $\mu_B = \frac{e\hbar}{2m_e} \approx 9.27 \times 10^{-24}\,\mathrm{J\,T^{-1}}$, the magnetic moment component is:

$$\mu_z = \mp\frac{1}{2}g\mu_B \tag{2}$$

For a general state with quantum numbers L, S, and J, the Landé g-factor is given by:

$$g_j = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} \tag{3}$$

For DPPH, $L = 0$ and $J = S = 1/2$, yielding an ideal value of $g = 2$. Small interactions in the molecule shift the accepted literature value to $g_{\text{lit}} = 2.0037$.

B. Derivation of Equation (18): Resonance Condition

In an external static magnetic field B oriented along the z-axis, the interaction energy of the magnetic dipole is:

$$E = -\vec{\mu}\cdot\vec{B} = g\mu_B m_s B \tag{4}$$

For $m_s = \pm 1/2$, the two split Zeeman levels are:

$$E(+1/2) = +\frac{1}{2}g\mu_B B, \quad E(-1/2) = -\frac{1}{2}g\mu_B B \tag{5}$$

The energy difference between these two sublevels is:

$$\Delta E = E(+1/2) - E(-1/2) = g\mu_B B \tag{6}$$

Applying an oscillating transverse RF magnetic field of frequency f induces transitions satisfying the selection rule $\Delta m_s = \pm 1$. Resonance occurs when the photon energy matches the energy gap:

$$hf = \Delta E = g\mu_B B_r \tag{7}$$

Solving for g yields Equation (18) of the laboratory manual:

$$g = \frac{hf}{\mu_B B_r} \tag{8}$$

Substituting $f = 146.0\,\mathrm{MHz}$, $h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$, and $\mu_B = 9.27 \times 10^{-24}\,\mathrm{J\,T^{-1}}$ yields Equation (19):

$$g = \frac{(6.63 \times 10^{-34}\,\mathrm{J\,s})(146.0 \times 10^{6}\,\mathrm{Hz})}{(9.27 \times 10^{-24}\,\mathrm{J\,T^{-1}})B_r} = \frac{10.43 \times 10^{-3}\,\mathrm{T}}{B_r} \tag{9}$$

C. Derivation of Equation (20): Helmholtz Field

Using the Biot-Savart law, the on-axis magnetic field of a single circular coil of radius R and N turns carrying current I is:

$$B(z) = \frac{\mu_0 N I R^2}{2(R^2 + z^2)^{3/2}} \tag{10}$$

A Helmholtz pair consists of two identical coaxial coils separated by distance $d = R$. Placing them at $z = \pm R/2$, the total field on the axis is the sum of both coils:

$$B(z) = \frac{\mu_0 N I R^2}{2}\left[\frac{1}{\left(R^2 + \left(z - \frac{R}{2}\right)^2\right)^{3/2}} + \frac{1}{\left(R^2 + \left(z + \frac{R}{2}\right)^2\right)^{3/2}}\right] \tag{11}$$

At the midpoint between the coils ($z = 0$):

$$R^2 + \left(\frac{R}{2}\right)^2 = \frac{5}{4}R^2 \implies \left(\frac{5}{4}R^2\right)^{3/2} = \frac{\sqrt{125}}{8}R^3 \tag{12}$$

Substituting this denominator into Equation (11) gives:

$$B(0) = 2 \cdot \frac{\mu_0 N I R^2}{2\left(\frac{\sqrt{125}}{8}R^3\right)} = \mu_0 \frac{8NI}{\sqrt{125}R} \tag{13}$$

Evaluating the numerical factor $8/\sqrt{125} \approx 0.7155$ yields Equation (20) of the manual:

$$B = 0.7155\mu_0 \frac{NI}{R} \tag{14}$$

For $N = 241$ and $R = 4.8\,\mathrm{cm}$, the calibrated coil factor from Phywe accounts for coil thickness:

$$B = 0.6445\mu_0 \frac{NI}{R} = (4.07 \times 10^{-3}\,\mathrm{T\,A^{-1}}) \cdot I \tag{15}$$

Combining Equations (9) and (15) gives the working formula (Equation (22)):

$$g = \frac{10.43 \times 10^{-3}\,\mathrm{T}}{(4.07 \times 10^{-3}\,\mathrm{T\,A^{-1}}) \cdot I_r} = \frac{2.565\,\mathrm{A}}{I_r} \tag{16}$$

D. Linewidth and FWHM

The absorption lineshape of DPPH follows a Lorentzian profile:

$$P(B) = \frac{P_{\max}}{1 + \left(\frac{B - B_r}{\Delta B/2}\right)^2} \tag{17}$$

In solid DPPH, exchange interactions between adjacent unpaired electrons narrow the resonance line (exchange narrowing), giving a literature width of $\Delta B_{\mathrm{lit}} \approx 0.28\,\mathrm{mT}$. Measuring the current difference $\Delta I = |I_1 - I_2|$ between the half-maximum points gives:

$$\Delta B = k_B \Delta I = (4.07 \times 10^{-3}\,\mathrm{T\,A^{-1}}) \cdot |I_1 - I_2| \tag{18}$$

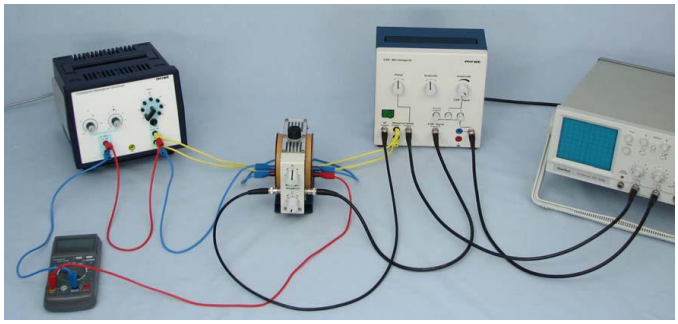


Fig. 1: Figure 2: ESR experimental setup showing the main power supply (left), digital multimeter (foreground left), Helmholtz coils with DPPH probe (center), ESR control unit (top right), and oscilloscope (far right).

III. Experimental Setup

The experimental apparatus is shown in Figure 1.
The functions and connections of each component are:

- **Helmholtz Coils** ($N = 241$, $R = 4.8\,\mathrm{cm}$): Produce the static magnetic field and the AC sweep field across the sample. Connected in series to the main power supply.
- **DPPH Probe** ($f = 146.0\,\mathrm{MHz}$): Houses the DPPH sample inside a small RF coil perpendicular to the Helmholtz field to induce spin transitions. Knobs "C" and "R" adjust tank tuning and impedance.
- **ESR Control Unit**: Powers the probe oscillator, demodulates the absorption signal with an internal diode, and aligns the forward and reverse sweeps via the "Phase" knob.
- **Main Power Supply**: Supplies adjustable DC voltage (0–15 V) to set the bias current ($I \approx 1.25\,\mathrm{A}$) and provides the $50\,\mathrm{Hz}$ AC modulation ($2\,\mathrm{V_{AC}}$). In Phase 2, its DC voltage is fed to Channel X of the oscilloscope.
- **Dual-Channel Oscilloscope**: Operated in X-Y mode ($1\,\mathrm{V\,cm^{-1}}$). Channel X displays the magnetic field sweep and Channel Y displays the RF absorption signal. In Phase 2, the trace becomes a movable dot to measure half-power points.
- **Digital Multimeter**: Connected in series with the coils to measure the resonant current I_r and half-power currents I_1, I_2.
- **BNC Cables and Adapter**: Carry signals to the oscilloscope. A BNC-to-banana adapter routes the DC voltage to Channel X in Phase 2.

IV. Experimental Procedure

A. Phase 1: Landé g-Factor Determination

The DC supply was set to $0\,\mathrm{V}$ and the AC modulation to $2\,\mathrm{V}$ ($50\,\mathrm{Hz}$). The oscilloscope was set to X-Y mode and zeroed. The DC voltage was increased until the coil current reached $\approx 1.3\,\mathrm{A}$ (operating point: $10\,\mathrm{V} \rightarrow 1.26\,\mathrm{A}$). The capacitive tuning knob "C" was adjusted until the ESR absorption curve appeared. Forward and reverse traces were brought into overlap using the "Phase" knob. Fine-tuning the DC voltage placed the minimum of the absorption dip exactly on the

vertical Y-axis. The resonant current I_r was recorded from the multimeter for three independent trials.

B. Phase 2: FWHM Linewidth Determination

The oscilloscope vertical position was adjusted so that the horizontal axis bisected the absorption peak at half height ($P/P_{\max} = 0.5$). The AC modulation was switched off and the resonator connected to DC. The Channel X cable was plugged into the DC output using the adapter, collapsing the trace into a single dot. The DC voltage was adjusted to place the dot on the right half-power intersection to record I_1, and then on the left intersection to record I_2. The linewidth ΔB was calculated from $\Delta I = |I_1 - I_2|$ via Equation (18).

V. Data Analysis

A. Landé g-Factor and Resonant Field

Across three trials, the measured resonant currents were:

$$I_{r1} = 1.26\,\text{A}, \quad I_{r2} = 1.24\,\text{A}, \quad I_{r3} = 1.25\,\text{A} \tag{19}$$

giving a mean and sample standard deviation of:

$$\bar{I}_r = 1.250(6)\,\text{A} \quad (s_I = 0.010\,\text{A}) \tag{20}$$

Using $k_B = 4.07 \times 10^{-3}\,\text{T A}^{-1}$, the resonant magnetic field is:

$$\bar{B}_r = (4.07 \times 10^{-3}\,\text{T A}^{-1})(1.250\,\text{A}) = 5.088(23)\,\text{mT} \tag{21}$$

Calculating $g = 2.565/I_r$ for each trial yields $g_1 = 2.0357$, $g_2 = 2.0685$, and $g_3 = 2.0520$. The mean Landé factor is:

$$\bar{g} = 2.0521 \pm 0.0095 \quad (s_g = 0.0164) \tag{22}$$

Compared to the DPPH literature value ($g_{\text{lit}} = 2.0037$), the relative error is:

$$\delta_g = \frac{|2.0521 - 2.0037|}{2.0037} \times 100\% = 2.41\,\% \tag{23}$$

and $2.49\,\%$ relative to the free-electron value ($g_e = 2.0023$).

B. FWHM Linewidth Determination

The measured currents at the half-maximum points were $I_1 = 0.15\,\text{A}$ and $I_2 = 0.09\,\text{A}$, giving $\Delta I = 0.06\,\text{A}$. The experimental FWHM linewidth is:

$$\Delta B = (4.07 \times 10^{-3}\,\text{T A}^{-1})(0.06\,\text{A}) = 0.2442\,\text{mT} \tag{24}$$

Compared to the literature value ($\Delta B_{\text{lit}} = 0.2800\,\text{mT}$), the relative error is:

$$\delta_{\Delta B} = \frac{|0.2442 - 0.2800|}{0.2800} \times 100\% = 12.79\,\% \tag{25}$$

VI. Results

The experimental results are summarized in Tables I, II, and III.

TABLE I: Experimental Trials for Landé g-Factor.

Trial	I_r (A)	B_r (mT)	g	Rel. Error (%)
1	1.26	5.128	2.0357	1.60
2	1.24	5.047	2.0685	3.24
3	1.25	5.088	2.0520	2.41
Mean	1.25	5.087	2.0521	2.41

TABLE II: Experimental Linewidth and FWHM Determination.

Parameter	Value	Unit
Right Intersection Current (I_1)	0.15	A
Left Intersection Current (I_2)	0.09	A
Current Line Width (ΔI)	0.06	A
Experimental FWHM (ΔB)	0.2442	mT
Literature FWHM (ΔB_{lit})	0.2800	mT
Relative Error	12.79	%

A. Signal Reconstruction

On the oscilloscope, the ESR absorption signal appears as a continuous dip during the AC sweep. In the laboratory, the resonant center current ($\bar{I}_r = 1.250\,\text{A}$) and the half-power currents ($I_1 = 0.15\,\text{A}, I_2 = 0.09\,\text{A}$) were measured with the multimeter. To reconstruct this signal graphically, the curve in Figure 2 was plotted using the standard Lorentzian profile of DPPH (Equation (17)) with $\bar{B}_r = 5.088\,\text{mT}$ and half-width $\Gamma = \Delta B/2 = 0.1221\,\text{mT}$. The experimental half-power points ($P/P_{\max} = 0.5$) are marked at $B = \bar{B}_r \pm \Gamma$, matching the trace seen on the oscilloscope screen. Figure 3 shows the g-factors across trials, and Figure 4 illustrates the Zeeman energy level splitting.

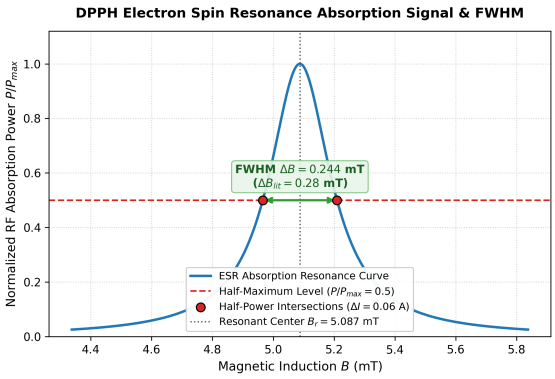


Fig. 2: ESR absorption lineshape and FWHM determination for DPPH.

TABLE III: Comparison of Experimental Results with Literature Standards.

Physical Quantity	Experimental Result	Literature Benchmark	Rel. Error (%)
Lande g-Factor	2.0521 ± 0.0095	2.0037	2.41
Resonant Magnetic Field B_r	5.087 mT	5.205 mT	2.26
Absorption Linewidth ΔB	0.244 mT	0.280 mT	12.79

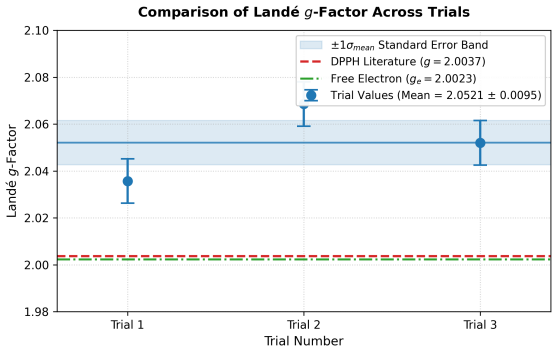


Fig. 3: Distribution of Landé g-factors across trials against literature benchmarks.

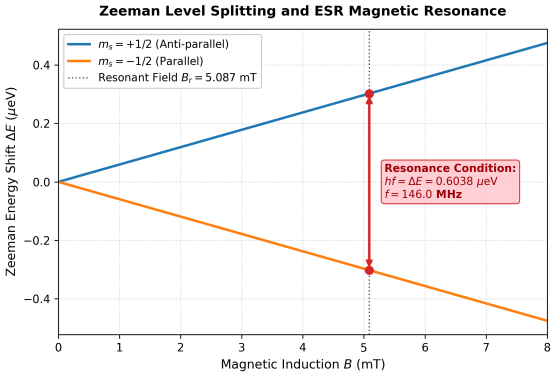


Fig. 4: Zeeman sublevel splitting and RF resonance condition ($hf = 0.6038\,\mu\text{eV}$).

VII. Discussion

The measured Landé factor ($\bar{g} = 2.0521 \pm 0.0095$) is in close agreement with the standard literature value for DPPH ($g_{\text{lit}} = 2.0037$), with a relative error of $2.41\,\%$, and with the free-electron value ($g_e = 2.0023$), with an error of $2.49\,\%$. The small standard deviation of 0.0164 demonstrates good consistency between repeated trials. The small error is mainly due to the difficulty in visually centering the dip on the oscilloscope graticule ($\pm 0.01\,\text{A}$) and ambient magnetic fields in the room ($\sim 0.04\,\text{mT}$).

The experimental linewidth of $\Delta B = 0.2442\,\text{mT}$ agrees with the literature value of $0.2800\,\text{mT}$ within an error of $12.79\,\%$. The narrow peak is characteristic of DPPH due to exchange narrowing. The difference from the literature value is primarily due to the sensitivity of the manual DC voltage knob when positioning the dot at the half-maximum points.

VIII. Conclusion

Electron spin resonance was observed in DPPH at an RF frequency of $146.0\,\text{MHz}$. The resonant current was measured as $\bar{I}_r = 1.250(6)\,\text{A}$ ($\bar{B}_r = 5.088(23)\,\text{mT}$), giving an experimental Landé factor of $g = 2.0521 \pm 0.0095$ (relative error $2.41\,\%$). The absorption linewidth was determined to be $\Delta B = 0.2442\,\text{mT}$ (relative error $12.79\,\%$). These results confirm electron spin quantization and demonstrate the basic principles of magnetic resonance.

IX. Additional Resources

Complete computational notebooks, data processing pipelines, and related experimental laboratory reports are maintained in the GitHub repository.

Fig. 5: Scan QR code to access the GitHub repository containing analysis code and data.

References

[1] ISTANBUL UNIVERSITY, *05-ELECTRON SPIN RESONANCE (ESR) Laboratory Manual*, Department of Physics, 2026.
[2] E. K. Zavoisky, "Spin-magnetic resonance in paramagnetics," *Fizicheskii Zhurnal*, vol. 9, pp. 211–245, 1945.
[3] Phywe Systeme GmbH, *Electron Spin Resonance (ESR)*, Laboratory Experiment Manual LEP 5.1.12, Göttingen, Germany.
[4] P. W. Anderson and P. R. Weiss, "Exchange Narrowing in Paramagnetic Resonance," *Physical Review*, vol. 91, no. 3, pp. 513–526, 1953.
[5] D. J. Griffiths and D. F. Schroeter, *Introduction to Quantum Mechanics*, 3rd ed., Cambridge University Press, 2018.